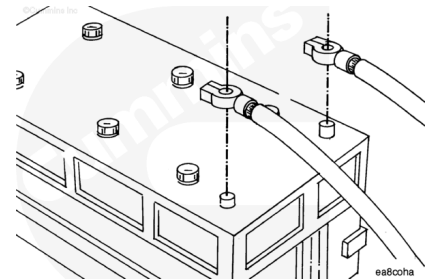


Trainings ▾

Resistance Check

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

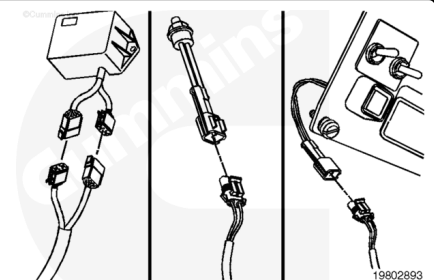


Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery. Disconnect the ICON™ battery harness connectors from the battery.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.



Disconnect the ICON™ idle control module A and B connectors.

Disconnect the parking brake switch from the cab harness.

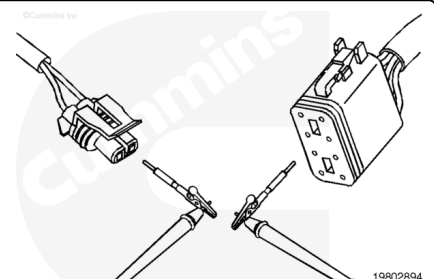
Disconnect the ICON™ lamp from the cab harness.

Set the multimeter to measure resistance.

Check the interlock output circuit.

Touch one of the multimeter leads to pin 5 of the ICON™ idle control module A harness connector. Touch the other multimeter lead to pin B of the parking brake switch harness connector.

Read the value displayed on the multimeter.



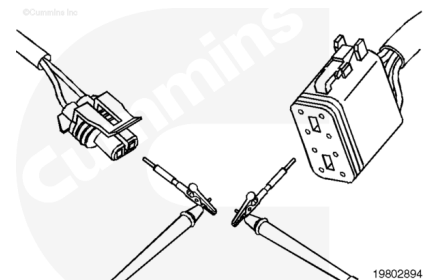
Remove the lead from pin B of the parking brake switch harness connector. Touch the multimeter lead to pin B of the ICON™ lamp harness connector.

Read the value displayed on the multimeter.

Finally, touch one of the multimeter leads to pin 6 of the ICON™ idle control module A harness connector.

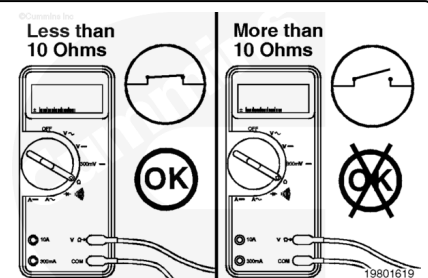
Touch the other multimeter lead to pin A of the ICON™ lamp harness connector.

Read the value displayed on the multimeter.



For all three pin checks, the multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If any of the circuits are **not** closed, isolate the problem to the cab harness or ICON™ engine harness side of the circuit. If the circuit is closed, check the interlock input circuit.



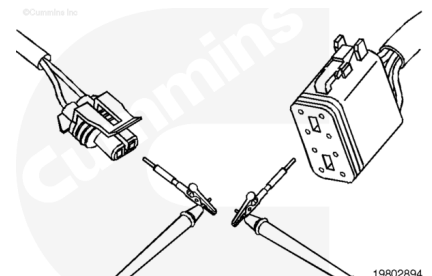
Check the interlock input circuit.

Make sure that the hood tilt and neutral position switches are closed.

Touch one of the multimeter leads to pin 3 of the ICON™ idle control module A harness connector.

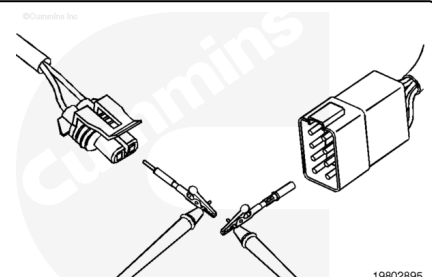
Touch the other multimeter lead to pin A of the parking brake switch harness connector.

Read the value displayed on the multimeter.



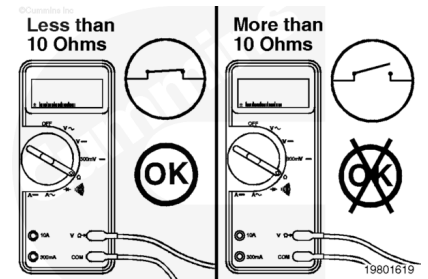
Then, with one lead still touching pin A of the parking brake switch harness connector, touch the other multimeter lead to pin 2 of the ICON™ idle control module B harness connector.

Read the value displayed on the multimeter.



For both pin checks, the multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If either circuit is **not** closed, isolate the problem to the cab harness or ICON™ engine harness side of the circuit.

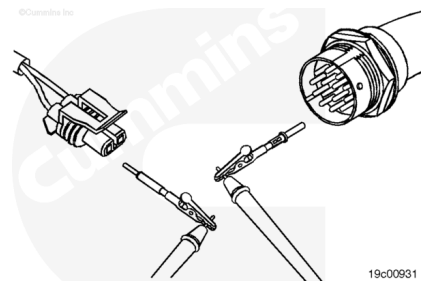


Check the output wire on the cab harness side.

Disconnect the 14-pin pass-through connector at the vehicle's firewall.

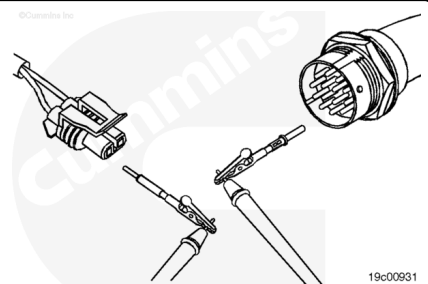
Touch one of the multimeter leads to pin J of the 14-pin pass-through connector, cab harness side. Touch the other multimeter lead to pin B of the parking brake switch harness connector.

Read the value displayed on the multimeter.



Then, with one lead still touching to pin J of the 14-pin connector, cab harness side, touch the second multimeter lead to pin B of the ICON™ lamp harness connector.

Read the value displayed on the multimeter.



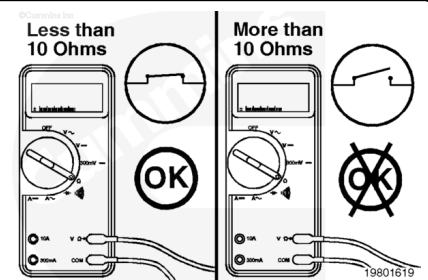
For both pin checks, the multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If either circuit is **not** closed, repair or replace the cab harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-200, or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

If both circuits are closed, repair or replace the ICON™ engine harness. Refer to Procedure 019-202

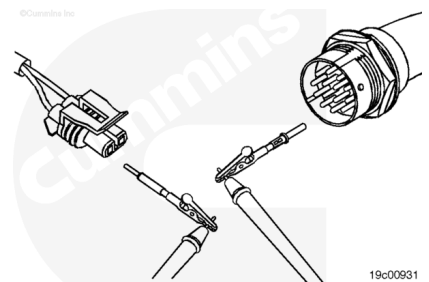
(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one multimeter lead to pin D of the 14-pin pass-through connector, cab harness side.

Touch the other multimeter lead to pin A of the ICON™ lamp harness connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

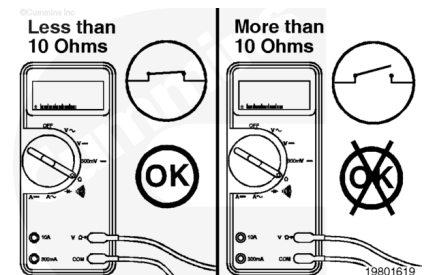
If the circuit is **not** closed, repair or replace the cab harness.

Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

If the circuit is closed, repair or replace the ICON™ engine harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

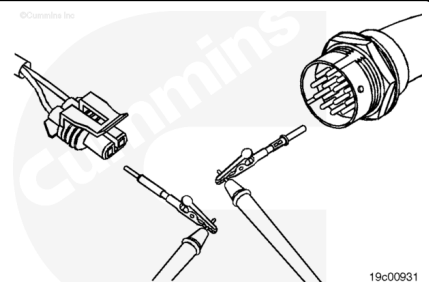


Check the input wire on the cab harness side.

Touch one of the multimeter leads to pin K of the 14-pin pass-through connector, cab harness side.

Touch the other multimeter lead to pin A of the parking brake switch harness connector.

Read the value displayed on the multimeter.



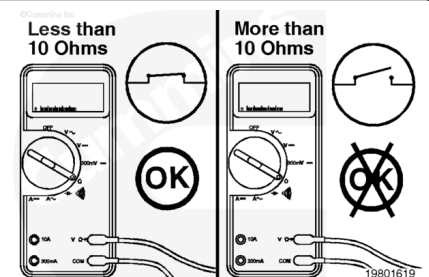
The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the cab harness.

Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

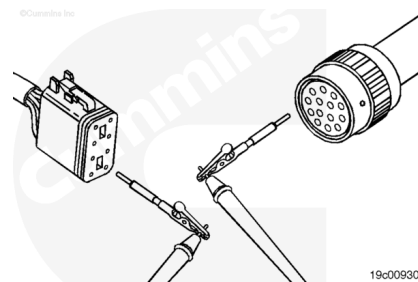
If the circuit is closed, isolate the problem to the correct portion of the ICON™ engine harness.



Touch one of the multimeter leads to pin K of the 14-pin pass-through connector, engine harness side.

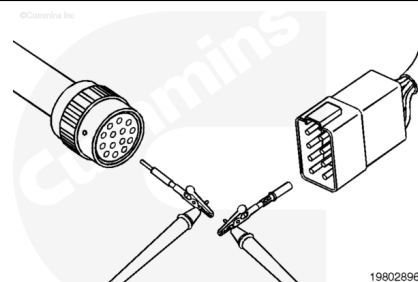
Touch the other multimeter lead to pin 3 of the ICON™ idle control module A harness connector.

Read the value displayed on the multimeter.



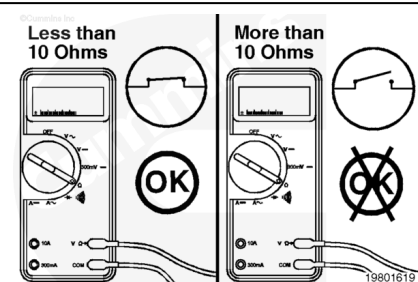
Then, leaving the first lead in place on pin K of the 14-pin connector, touch the second multimeter lead to pin 2 of the ICON™ idle control module B harness connector.

Read the value displayed on the multimeter.



For both pin checks, the multimeter **must** display a reading of less than 10 ohms (closed circuit).

If the circuit is **not** closed, check the portion of the circuit to the neutral position switch.

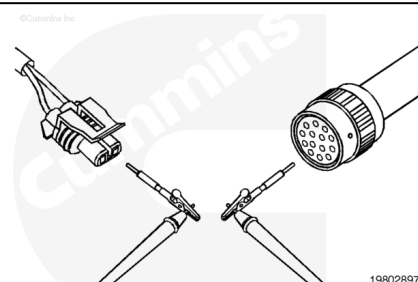


First, verify that the neutral position switch is good. Replace neutral position switch as required.

Disconnect the neutral position switch from the ICON™ engine harness.

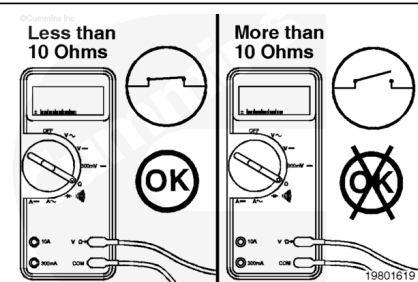
Touch one of the multimeter leads to pin K of the 14-pin pass-through connector, engine harness side. Touch the other multimeter lead to pin B of the neutral position switch harness connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 10 ohms (closed circuit).

If the circuit is **not** closed, repair or replace the ICON™ engine harness.



Refer to Procedure 019-202

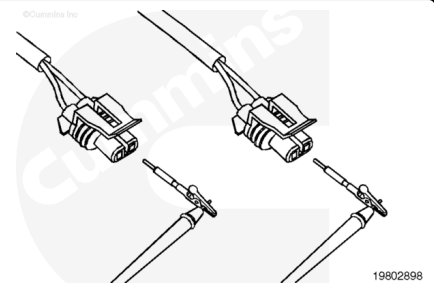
(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

Disconnect the hood tilt switch from the ICON™ engine harness.

Touch one of the multimeter leads to pin A of the neutral position switch harness connector.

Touch the other multimeter lead to pin B of the hood tilt switch harness connector.

Read the value displayed on the multimeter.

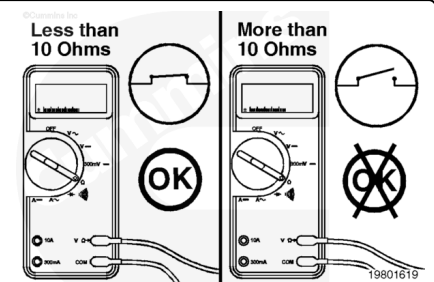


The multimeter **must** display a reading of less than 10 ohms (closed circuit).

If the circuit is **not** closed, repair or replace the ICON™ engine harness.

Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

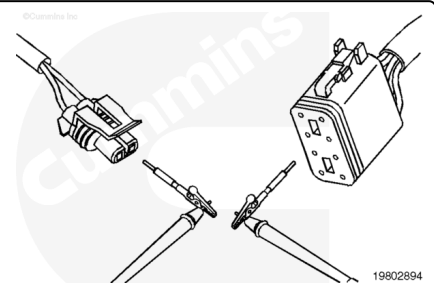


Verify that the hood tilt switch is good.

Touch one of the multimeter leads to pin A of the hood tilt switch harness connector.

Touch the other multimeter lead to pin 3 of the ICON™ idle control module A harness connector.

Read the value displayed on the multimeter.



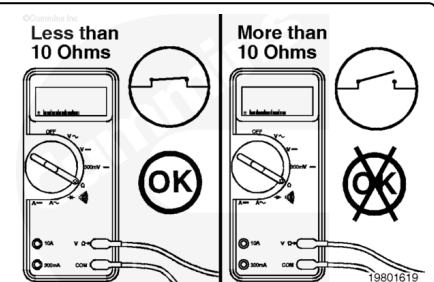
The multimeter **must** display a reading of less than 10 ohms (closed circuit).

If the circuit is **not** closed, repair or replace the ICON™ engine harness.

Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

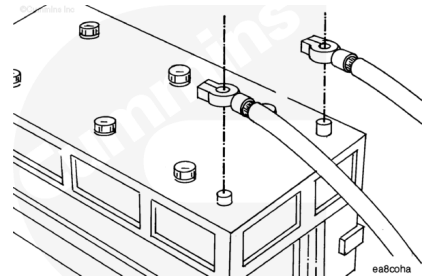
Connect all components after completing the repair.



Check for Short Circuit to Ground

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

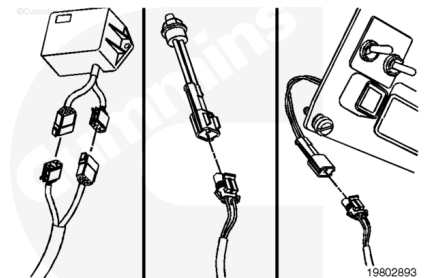


Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery. Disconnect the ICON™ battery harness connectors from the battery.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part No 3822758 and 3822917, when taking a measurement.



Disconnect the ICON™ idle control module A and B connectors.

Disconnect the 14-pin pass-through connector at the vehicle's firewall.

Disconnect the parking brake switch and the ICON™ lamp from the cab harness.

Disconnect the neutral position switch and hood tilt switch from the ICON™ engine harness.

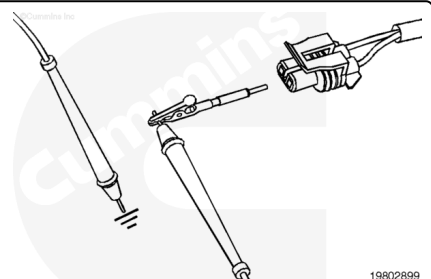
Set the multimeter to measure resistance.

Touch one of the multimeter leads to pin B of the ICON™ lamp harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin A of the ICON™ lamp harness connector. Touch the other multimeter lead to ground.

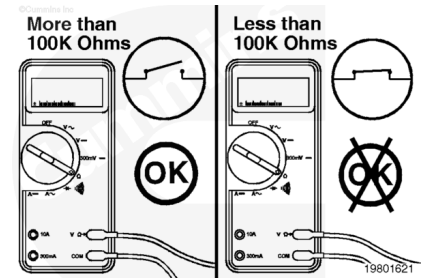
Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground.

Isolate the short to the cab harness or engine harness portion of the circuit.

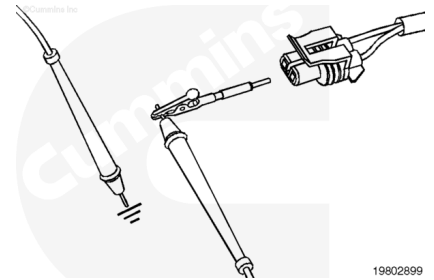


Touch one of the multimeter leads to pin B of the parking brake switch harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin A of the parking brake switch harness connector. Touch the other multimeter lead to ground.

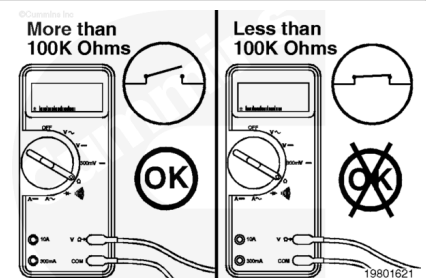
Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

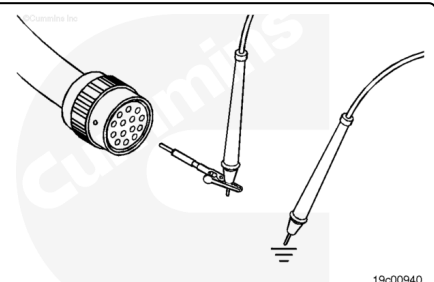
If the circuit is **not** open, there is a short circuit to ground.

Isolate the short to the cab harness or engine harness portion of the circuit.

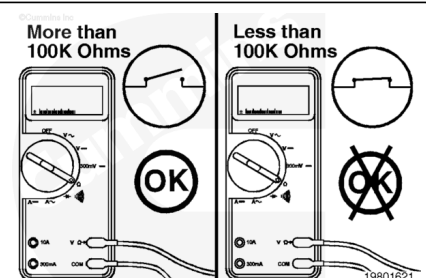


Touch one of the multimeter leads to pin J of the 14-pin pass-through connector, engine harness side. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin D of the 14-pin pass-through connector, engine harness side. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If either circuit is **not** open, check for a short circuit from the appropriate idle control module harness connector pin.



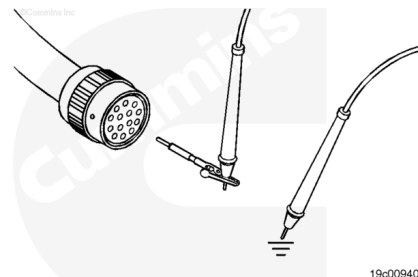
If either circuit is open, repair or replace the cab harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

Touch one of the multimeter leads to pin K of the 14-pin pass-through connector, engine harness side.

Touch the other multimeter lead to ground.

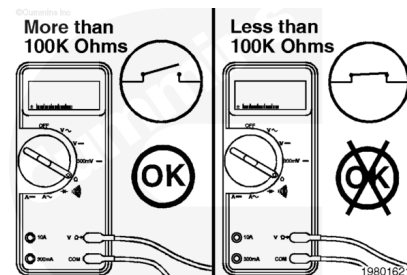
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, isolate the short to either idle control module connector.

If the circuit is open, repair or replace the cab harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-200

(/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

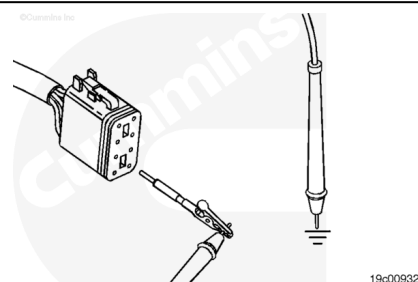


Touch one of the multimeter leads to pin 5 of the ICON™ idle control module A harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin 6 of the ICON™ idle control module A harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

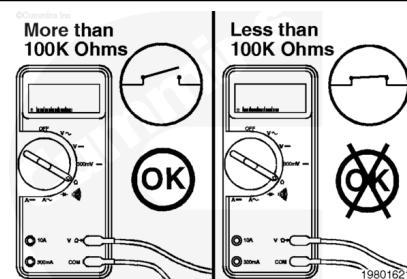


For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If either circuit is **not** open, repair or replace the ICON™ engine harness.

Refer to Procedure 019-208

(/qs3/pubsys2/xml/en/procedures/97/97-019-208.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-

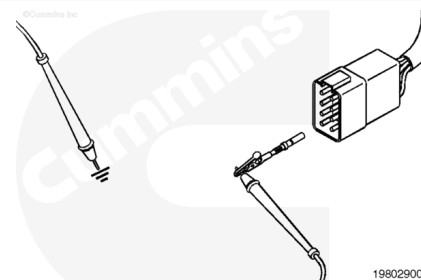


043.html).

Touch one of the multimeter leads to pin 2 of the ICON™ idle control module B harness connector.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

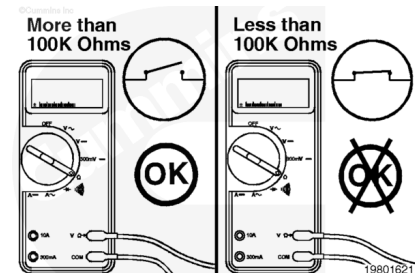


The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, repair or replace the ICON™ engine harness.

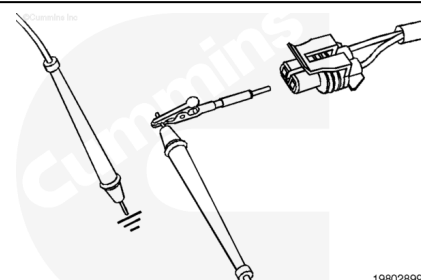
Refer to Procedure 019-208

(/qs3/pubsys2/xml/en/procedures/97/97-019-208.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin A of the neutral position switch harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.

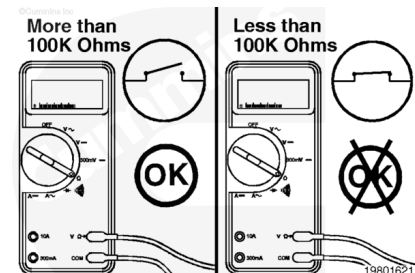
Then, touch one of the multimeter leads to pin B of the neutral position switch harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

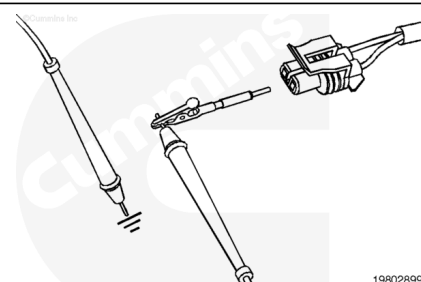
If either circuit is **not** open, repair or replace the ICON™ engine harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin A of the hood tilt switch harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.

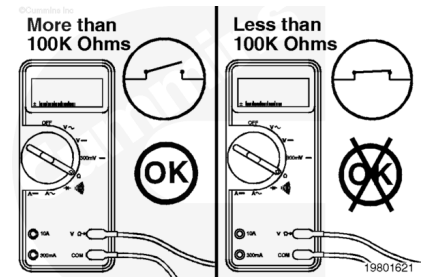
Then, touch one of the multimeter leads to pin B of the hood tilt switch harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If either check is **not** open, repair or replace the ICON™ engine harness. Refer to Procedure 019-202

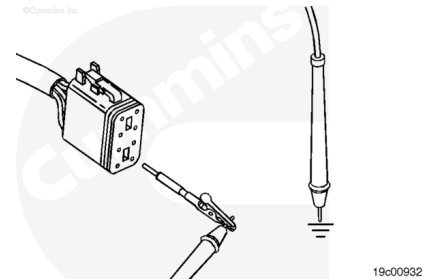
(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

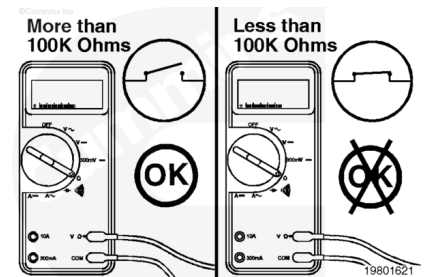


The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If either circuit is **not** open, repair or replace the ICON™ engine harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

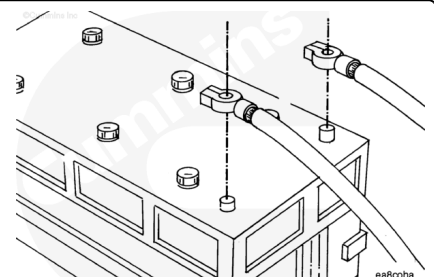
Connect all components after completing the repair.



Check for Short Circuit from Pin to Pin

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery. Disconnect the ICON™ battery harness connectors from the battery.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

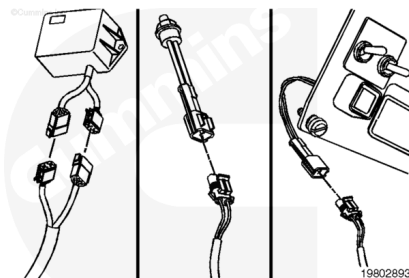
To avoid pin and harness damage, use test leads, Part No 3822758 and 3822917, when taking a measurement.

Disconnect the ICON™ idle control module A and B connectors. Disconnect the 14-pin pass-through connector at the vehicle's firewall.

Disconnect the parking brake switch and the ICON™ lamp from the cab harness.

Disconnect the neutral position switch and the hood tilt switch from the ICON™ engine harness.

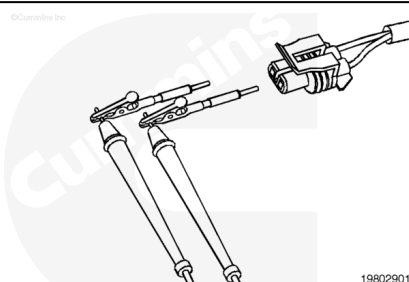
Set the multimeter to measure resistance.



Touch one of the multimeter leads to pin A of the parking brake switch harness connector.

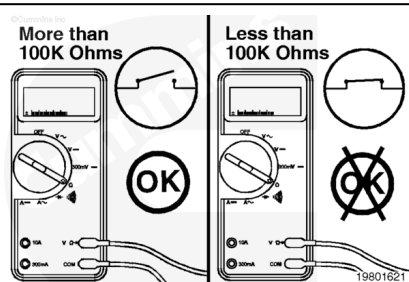
Touch the other multimeter lead to pin B of the harness connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit between pin A and pin B.

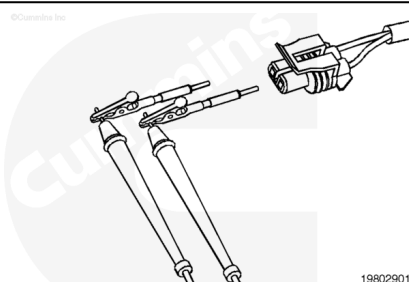
Repair or replace the cab harness. Refer to Procedure 019-202 (</qs3/pubsys2/xml/en/procedures/97/97-019-202.html>) or 019-305 (</qs3/pubsys2/xml/en/procedures/97/97-019-305.html>).



Touch one of the multimeter leads to pin A of the ICON™ lamp harness connector.

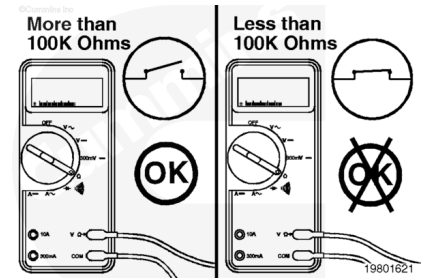
Touch the other multimeter lead to pin B of the harness connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit between pin A and pin B.

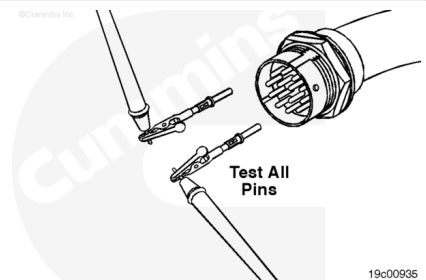
Repair or replace the cab harness. Refer to Procedure 019-202 (</qs3/pubsys2/xml/en/procedures/97/97-019-202.html>) or 019-305 (</qs3/pubsys2/xml/en/procedures/97/97-019-305.html>).



Touch one of the multimeter leads to pin J of the 14-pin pass-through connector, cab harness side. Touch the other multimeter lead to all other pins in the connector, one at a time.

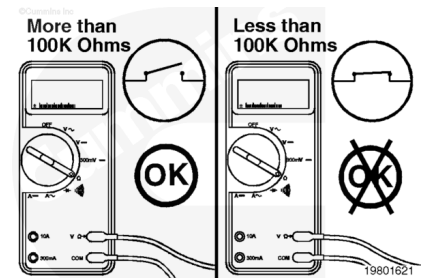
Repeat the pin-to-pin check from pin D of the 14-pin connector, cab harness side, to all other pins in the connector, and from pin K to all other pins.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit from pin J, pin D, or pin K to any other pin in the connector that registered a closed circuit.

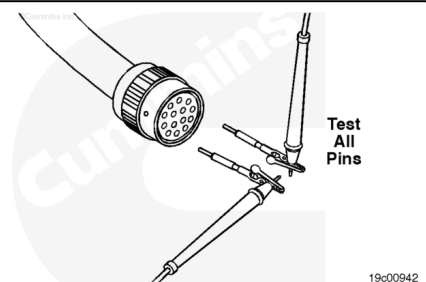
Repair or replace the cab harness. Refer to Procedure 019-200 (</qs3/pubsys2/xml/en/procedures/97/97-019-200.html>) or 019-305 (</qs3/pubsys2/xml/en/procedures/97/97-019-305.html>).



Touch one of the multimeter leads to pin J of the 14-pin pass-through connector, engine harness side. Touch the other multimeter lead to all other pins in the connector, one at a time.

Repeat the pin-to-pin check from pin D of the 14-pin connector, engine harness side, to all other pins in the connector, and from pin K to all other pins.

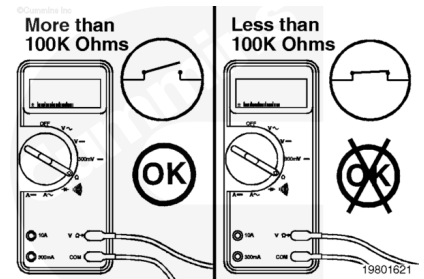
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin J, pin D, or pin K in the 14-pin pass-through connector to any other pin in the connector that registered a closed circuit.

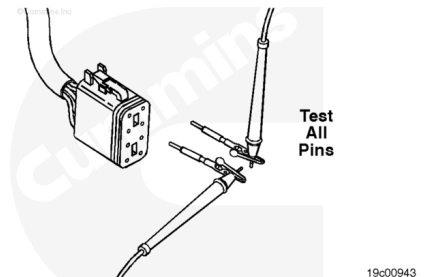
Repair or replace the ICON™ engine harness. Refer to Procedure 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module A harness connector. Touch the other multimeter lead to all other pins in the connector, one at a time.

Repeat the pin-to-pin check from pin 5 of the ICON™ idle control module A harness connector to all other pins in the connector, and from pin 6 to all other pins.

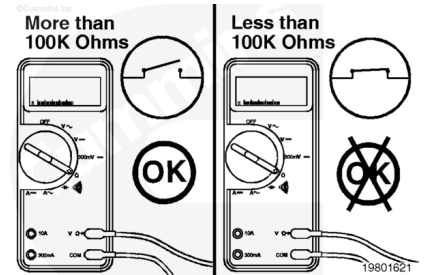
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin 3, pin 5, or pin 6 in the ICON™ idle control module A connector to any other pin in the connector that registered a closed circuit.

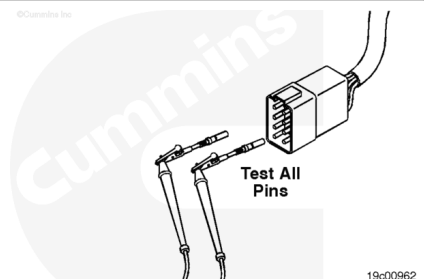
Repair or replace the ICON™ engine harness. Refer to Procedure 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 2 of the ICON™ idle control module B harness connector.

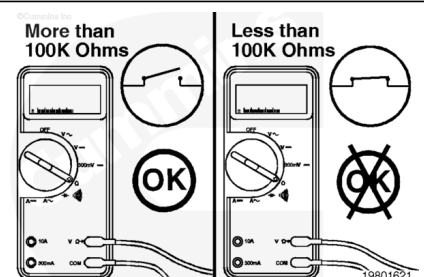
Touch the other multimeter lead to all other pins in the connector, one at a time.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin 2 in the ICON™ idle control module B connector to any other pin in the connector that registered a closed circuit.

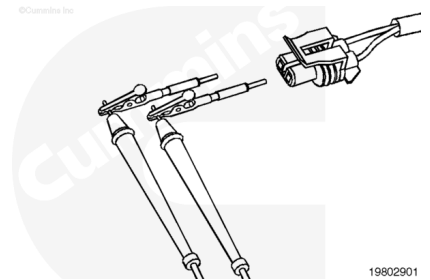


Repair or replace the ICON™ engine harness. Refer to Procedure 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

Touch one of the multimeter leads to pin A of the neutral position switch harness connector.

Touch the other multimeter lead to pin B of the connector.

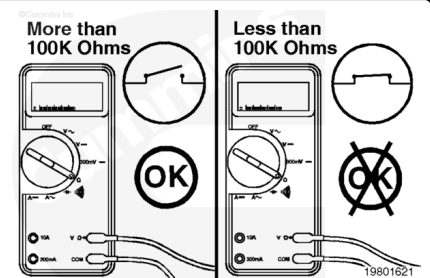
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin A to pin B in the neutral position switch harness connector.

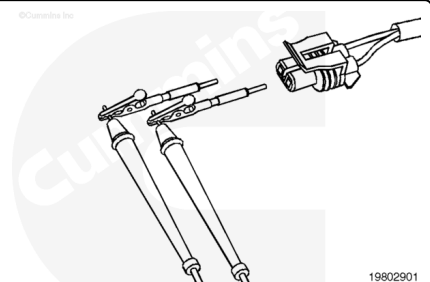
Repair or replace the ICON™ engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin A of the hood tilt switch harness connector.

Touch the other multimeter lead to pin B of the connector.

Read the value displayed on the multimeter.

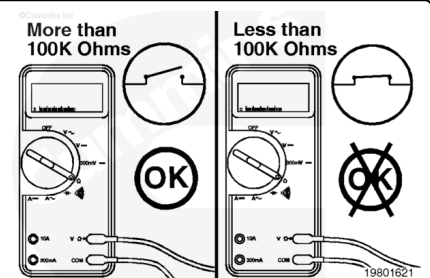


The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin A to pin B in the hood tilt switch harness connector.

Repair or replace the ICON™ engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/97/97-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

Connect all components after completing the repair.



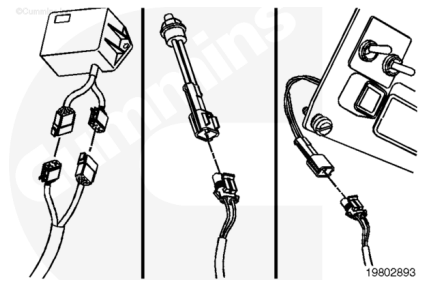
Check for Short Circuit to External Voltage Source

Disconnect the ICON™ idle control module A and B connectors.

Disconnect the ICON™ lamp and the parking brake switch from the cab harness.

Set the multimeter to measure VDC.

Turn the keyswitch to the ON position.



⚠ CAUTION ⚠

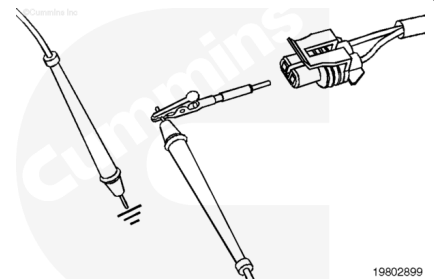
To avoid pin and harness damage, use test lead, Part Number 3822758, when taking a measurement.

Touch one of the multimeter leads to pin A of the ICON™ lamp harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin B of the ICON™ lamp harness connector. Touch the other multimeter lead to ground.

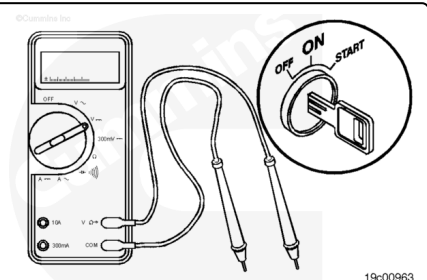
Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab or engine harness wiring that carries voltage.

Remove the external voltage source.

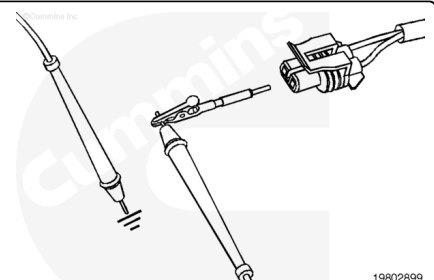


Touch one of the multimeter leads to pin A of the parking brake switch harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin B of the parking brake switch harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



For both pin-to-ground checks, the multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab or engine harness wiring that carries voltage.

Remove the external voltage source.

Connect all components after completing the repair.

